

Stimulus Repetitions in Whole Report

Experiment

In the present experiment, the effects of letter repetitions in whole report were tested in several different ways comparing report from displays with and without letter repetition: (1) the mean correct reports, intrusions (reported letter repetitions not present in the display), and error reports were measured across different exposure durations; (2) the probability of correct report of individual letters was explored as a function of the number of repetitions and exposure durations; and (3) the probability of correct report of individual letters was investigated as a function of the number of identical neighbors and exposure duration.

Methods

Participants. Ten undergraduate students (3 female) from the University of California, Santa Barbara participated in the experiment. The students received course credits for their participation and were naïve with respect to the purpose of the experiment. All participants had normal or correct-to-normal visual acuity. The experiment was approved by the local ethical committee of the University of California, Santa Barbara.

Stimulus material. The stimulus material consisted of uppercase letters printed in Arial font and colored in red. The mean width of the letters was 1.05 degrees of visual angle (range .43–1.53) and the height was 1.03 degrees of visual angle at a viewing distance of 100 cm. Six stimulus positions were used located equally spaced around the perimeter of an imaginary circle centered at fixation with a radius of 3.91 degrees of visual angle. Pattern masks consisting of red and blue simple geometric shapes were used to terminate stimulus processing. The width and height of the masks were 1.94 degrees of visual angle. A white fixation cross was continuously visible in the center of the screen. All stimuli were presented on a black background.

Procedure. The experiment was run on a CRT running at 75 Hz controlled by a PC using Matlab and the Psychophysics Toolbox extensions (Brainard, 1997). Each trial began with the presentation of the fixation cross in the center of the screen. When properly fixated, the participant initiated the presentation of six letters presented at the six stimulus positions. After the presentation of the stimulus letters, six pattern masks were presented at the six stimulus positions. Six different exposure durations were used: 13, 27, 40, 53, 93, and 200 ms. The exposure duration of the masks was 500 ms.

The task of the participants was to report as many of the six letters as possible. After the offset of the masks, the participant typed in the identity of the remembered letters in any order that they preferred. Participants were informed that reaction time was not recorded and that they could use as much time as they wished for responding. Before each experimental block, the participants were informed whether letter repetitions could occur or not in that particular block. In blocks where repetitions were possible, the participants were asked to type in repeating letters as many times as they recognized them in the display. Thus, if the participant recognized and remembered the letters *KHLH* from a display, the participant was asked to type the letter *H* twice in their report.

Design. Each participant ran a total of four experimental blocks in a single experimental session. Two different types of blocks were run: (A) letters drawn without replacement (WoR), and (B) letters drawn with replacement (WiR). In the WoR blocks (type *A*), six different letters were drawn from the set of 26 letters of the English alphabet in each trial without replacement so that all six letters were different. In the WiR blocks (type *B*), six different letters from the set of 26 letters of the English alphabet were first drawn. From this set, six letters were drawn WiR thus comprising the set of six letters for that particular trial. Half the participants ran the four blocks in the order *ABBA* and the other half in the order *BAAB* to counterbalance practice effects.

Each block of trials comprised 20 trials in each of the six exposure duration, yielding a total of 480 trials per participant (4 x 6 x 20 repetitions). The trial order was randomized within each experimental block. The participants were familiarized with the stimulus procedure in two practice trial blocks.

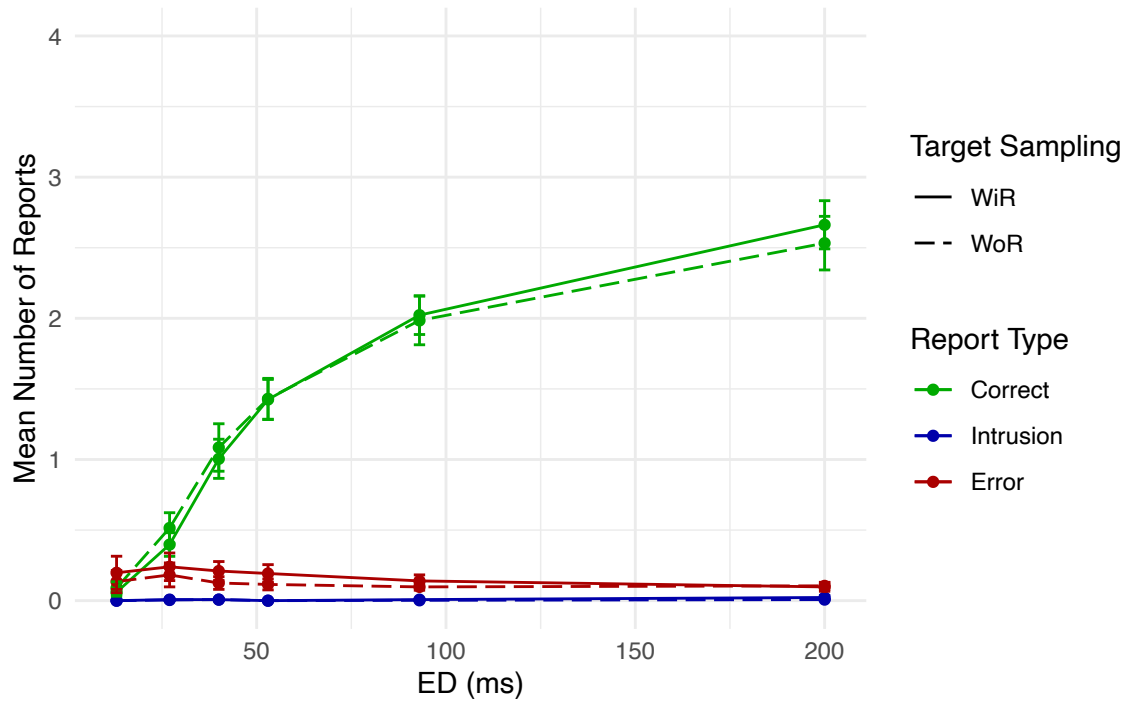


Figure 2. Average number of letters reported correctly (green), intrusions (blue), and errors (red) across the ten participants as a function of the different types of experimental blocks with letter selection WiR (solid lines) and WoR (dashed lines) and exposure duration. Intrusions are reports of letter repetitions that were not present in the display. Errors are report of letters not in the display. Error bars show the standard error of the means.

Results

I calculated the probability of a correct report of the individual letters in the display, the number of intrusions (reported letter repetitions *not* in the display), and the number errors in the following way: (1) calculated the ratio between the number of reported letters divided by the number of presented repetitions for each type of letter (but with a maximum of 1.0); (2) designated this ratio to each of the locations where the type of letter had been presented; (3) the number of correctly reported letters was then calculated as the sum of the ratios for all six locations; (4) if the number reports of a letter type was higher than the number of displayed letters of that type, the count of intrusions

was increased accordingly; finally (5) any letters reported that were not in the display were counted as errors.

Figure 2 shows the overall results of the experiment. The figure shows the mean number of correctly reported letters, intrusions, and errors as a function of the two different types of blocks (WiR and WoR) and exposure duration. Similar to previous results (e.g., Shibuya & Bundesen, 1988), mean correct reports were close to zero for the shortest exposure duration and rose towards the capacity of visual short-term memory (VSTM) of about three letters at the longest exposure duration of 200 ms. Error rates were highest for short exposure durations and then decreased. Intrusions were practically absent.

I fitted a linear mixed model to the data with Letter Selection (WiR or WoR), Response Type (correct, error, or intrusions), and Exposure Duration as fixed effects factors and Participants as the only random effect, intercept only (Baayen et al., 2008), using the R packages lme4 1.1.23 by Bates et al. (2014); lmerTest 3.1.3 by Kunzetsova et al. (2017); and effectsize 0.4.5 by Ben-Shachar et al. (2020). Surprisingly, the main effect of Letter Selection was insignificant, $F < 1$. The main effects of both Response Type and Exposure Duration were both significant, $F(2,315) = 1029.98$, $p < .001$, $\eta^2 = 0.87$ [CI 0.84;0.89] and $F(5,315) = 100.66$, $p < .001$, $\eta^2 = 0.62$ [0.55;0.66], respectively. In addition, only the two way interaction between Response Type and Exposure Duration was significant, $F(10,315) = 111.03$, $p < .001$, $\eta^2 = 0.78$ [0.74;0.81]. Thus, in contrast to the findings of Wolford and Hollingsworth (1974), there were no effect of letter repetitions in the present whole report experiment.

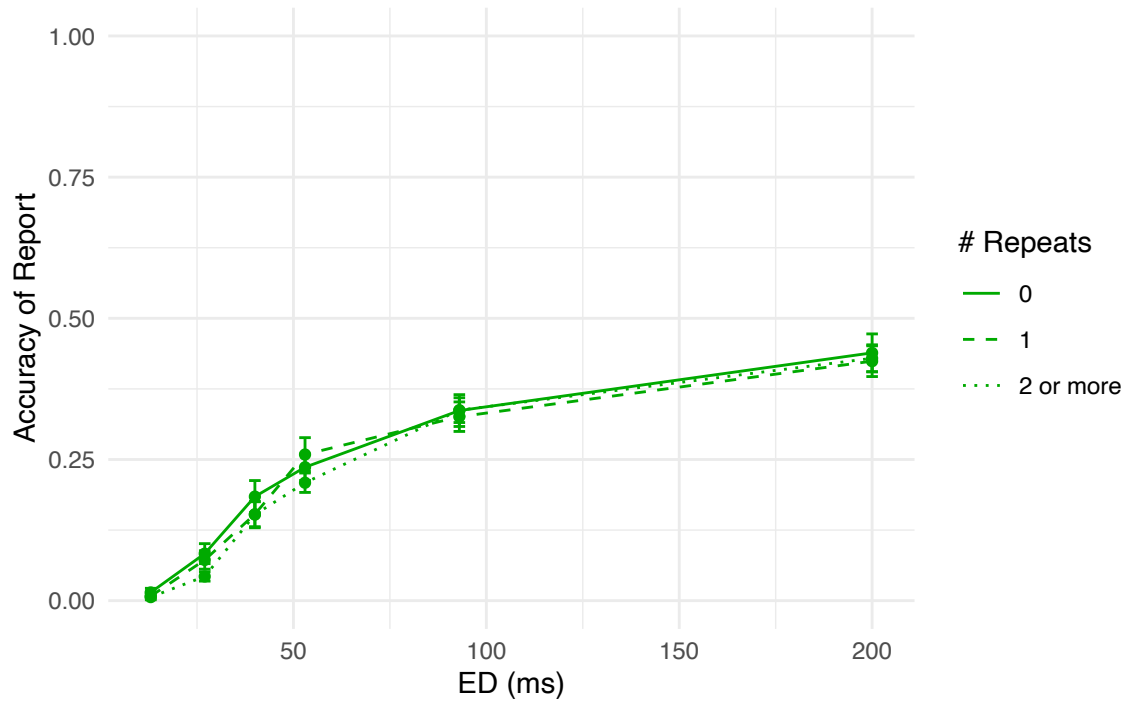


Figure 3. The probability of reporting a letter correctly as a function of exposure duration and the number of repetitions of letters within the displays. Error bars show the standard error of the means.

The lack of effect of letter repetitions in the first analyses of the data may have been due to two opposing effects. In the WiR displays, fewer reported repeated letters may have been accompanied with an increase in the number of non-identical letters reported. Consequently, the two effects could have cancelled each other out. To further explore this possibility, I analyzed the probability of reporting each of the individual six letters in the displays as a function of the number of repetitions. In Figure 3, the mean probabilities of correct reports are plotted as a function of the number of repetitions and exposure duration.

I fitted a linear mixed model with number of Repetitions (0, 1, or 2 or more) and Exposure Duration as fixed effect factors and Participants as random effect. The main effect of Exposure Duration, $F(5,159) = 101.94, p < .001, \eta^2 = 0.76 [0.7;0.81]$ was significant, but the effect of Repetitions was not, $F(2,153) = 1.97, p = .14, \eta^2 = 0.03$

[0;0.08]. The interaction between the two factors was also non significant, $F < 1$. Thus, again I found no effect of letter repetitions.

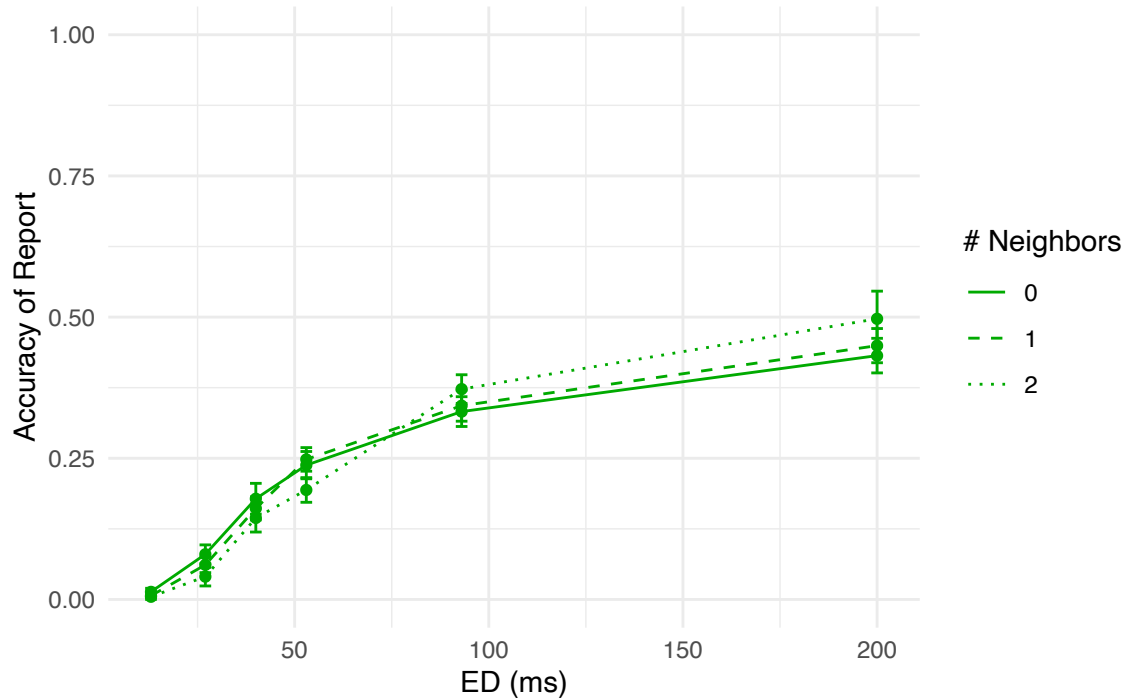


Figure 4. The probability of reporting a letter correctly as a function of exposure duration and the number of repetitions presented at neighboring display locations. Error bars show the standard error of the means.

I then continued to test the second hypothesis, that the probability of reporting identical letters at neighboring display locations is higher than for non-identical neighboring letters. Figure 4 shows the data for correct reports of letters with 0, 1, or 2 neighbors, respectively. I fitted a linear mixed model with Number of Neighbors and Exposure Duration as fixed effect factors and Participants as random effect. The main effect of Neighbors was not significant, $F < 1$, but the effect of Exposure Duration was, $F(5,152) = 208.38, p < .001, \eta^2 = 0.87 [0.84;0.9]$. The interaction between the two factors was not significant, $F(10,152) = 1.50, p = .15, \eta^2 = 0.09 [0;0.13]$. Looking at the plot in Figure 4, there is an indication of a small positive effect when a letter is surrounded by

two identical neighbors. To test this, I ran Bonferroni corrected pairwise comparisons conditioned on exposure duration, comparing the effect of the number of neighbors within each exposure duration. The analyses yielded a trend towards a significant difference when there were no and two identical neighbors at the longest exposure duration ($p = .08$), i.e., performance being higher in the later condition. All other comparisons had p -values larger than .15.